SQL- ASSIGNMENT-5 (Customer_Orders_Products Database)

Task 1 :-

1. Write a query to retrieve all records from the Customers table.

- select * from Customers;
- Output:

	CustomerID	Name	Email
1	1	John Doe	johndoe@example.com
2	2	Jane Smith	janesmith@example.com
3	3	Robert Johnson	robertjohnson@example.com
4	4	Emily Brown	emilybrown@example.com
5	5	Michael Davis	michaeldavis@example.com
6	6	Sarah Wilson	sarahwilson@example.com
7	7	David Thompson	davidthompson@example.com
8	8	Jessica Lee	jessicalee@example.com
9	9	William Turner	williamturner@example.com
10	10	Olivia Martinez	oliviamartinez@example.com
11	11	James Anderson	jamesanderson@example.com
12	12	Kelly Clarkson	kellyclarkson@example.com

2. Write a query to retrieve the names and email addresses of customers whose names start with 'J'.

- SELECT Name, Email FROM Customers WHERE Name LIKE 'J%';
- Output:

	Name	Email
1	John Doe	johndoe@example.com
2	Jane Smith	janesmith@example.com
3	Jessica Lee	jessicalee@example.com
4	James Anderson	jamesanderson@example.com

3. Write a query to retrieve the order details (OrderID, ProductName, Quantity) for all orders.

- SELECT OrderID, ProductName, Quantity FROM Orders;
- Output:

	OrderID	ProductName	Quantity
1	1	Product A	5
2	2	Product B	3
3	3	Product C	2
4	4	Product A	1
5	5	Product B	4
6	6	Product C	2
7	7	Product A	3
8	8	Product B	2
9	9	Product C	5
10	10	Product A	1
11	11	Product D	3
12	12	Product E	6
13	13	Product G	2
14	14	Product H	4
15	15	Product I	3

4.Write a query to calculate the total quantity of products ordered.

- SELECT SUM(Quantity) AS Total_Quantity FROM Orders;
- Output:

	Total_Quantity
1	46

5. Write a query to retrieve the names of customers who have placed an order.

- SELECT DISTINCT C.Name FROM Customers C INNER JOIN Orders O ON C.CustomerID = O.CustomerID;
- Output:

	Name
1	David Thompson
2	Emily Brown
3	James Anderson
4	Jane Smith
5	Jessica Lee
6	John Doe
7	Kelly Clarkson
8	Michael Davis
9	Olivia Martinez
10	Robert Johnson
11	Sarah Wilson
12	William Turner

6. Write a query to retrieve the products with a price greater than $10.00.

- SELECT * FROM Products WHERE Price > 10.00;
- Output:

	ProductID	ProductName	Price
1	1	Product A	10.99
2	4	Product D	12.99
3	8	Product H	11.99
4	9	Product I	14.99
5	12	Product L	15.99

7. Write a query to retrieve the customer name and order date for all orders placed on or after '2023-07-05'.

- SELECT C.Name, O.OrderDate
  FROM Customers C
  JOIN Orders O ON C.CustomerID = O.CustomerID
  WHERE O.OrderDate >= '2023-07-05';
- Output:

	Name	OrderDate
1	Michael Davis	2023-07-05
2	Sarah Wilson	2023-07-06
3	David Thompson	2023-07-07
4	Jessica Lee	2023-07-08
5	William Turner	2023-07-09
6	Olivia Martinez	2023-07-10
7	James Anderson	2023-07-10
8	Kelly Clarkson	2023-07-11
9	Michael Davis	2023-07-12
10	Emily Brown	2023-07-13
11	Sarah Wilson	2023-07-14

8. Write a query to calculate the average price of all products.

- SELECT AVG(Price) AS Average_Product_Price FROM Products;
- Output:

	Average_Product_Price
1	9.656666

9. Write a query to retrieve the customer names along with the total quantity of products they have ordered.

- SELECT C.Name, SUM(O.Quantity) AS Total_Quantity FROM Customers C JOIN Orders O ON C.CustomerID = O.CustomerID GROUP BY C.Name;
- Output:

	Name	Total_Quantity
1	David Thompson	3
2	Emily Brown	5
3	James Anderson	3
4	Jane Smith	3
5	Jessica Lee	2
6	John Doe	5
7	Kelly Clarkson	6
8	Michael Davis	6
9	Olivia Martinez	1
10	Robert Johnson	2
11	Sarah Wilson	5
12	William Turner	5

10. Write a query to retrieve the products that have not been ordered.

- SELECT P.ProductID, P.ProductName FROM Products P LEFT JOIN Orders O ON P.ProductName = O.ProductName WHERE O.ProductName IS NULL;
- Output:

	ProductID	ProductName
1	6	Product F
2	10	Product J
3	11	Product K
4	12	Product L

Task 2 :-

1. Write a query to retrieve the top 5 customers who have placed the highest total quantity of orders.

- SELECT TOP 5 C.Name, SUM(O.Quantity) AS Total_Quantity FROM Customers C JOIN Orders O ON C.CustomerID = O.CustomerID GROUP BY C.Name ORDER BY Total_Quantity DESC;
- Output:

	Name	Total_Quantity
1	Kelly Clarkson	6
2	Michael Davis	6
3	John Doe	5
4	Emily Brown	5
5	Sarah Wilson	5

2. Write a query to calculate the average price of products for each product category.

- SELECT ProductName, AVG(Price) AS Avg_Price FROM Products GROUP BY ProductName;
- Output:

	ProductName	Avg_Price
1	Product A	10.990000
2	Product B	8.990000
3	Product C	5.990000
4	Product D	12.990000
5	Product E	7.990000
6	Product F	6.990000
7	Product G	9.990000
8	Product H	11.990000
9	Product I	14.990000
10	Product J	4.990000
11	Product K	3.990000
12	Product L	15.990000

3. Write a query to retrieve the customers who have not placed any orders.

- SELECT C.CustomerID, C.Name FROM Customers C LEFT JOIN Orders O ON C.CustomerID = O.CustomerID WHERE O.CustomerID IS NULL;
- Output:

CustomerID	Name

4. Write a query to retrieve the order details (OrderID, ProductName, Quantity) for orders placed by customers whose names start with 'M'.

- SELECT O.OrderID, O.ProductName, O.Quantity FROM Orders O JOIN Customers C ON O.CustomerID = C.CustomerID WHERE C.Name LIKE 'M%';
- Output:

	OrderID	ProductName	Quantity
1	5	Product B	4
2	13	Product G	2

5. Write a query to calculate the total revenue generated from all orders.

- SELECT SUM(O.Quantity * P.Price) AS Total_Revenue FROM Orders O JOIN Products P ON O.ProductName = P.ProductName;
- Output:

	Total_Revenue
1	444.54

6. Write a query to retrieve the customer names along with the total revenue generated from their orders.

- SELECT C.Name, SUM(O.Quantity * P.Price) AS Revenue FROM Customers C JOIN Orders O ON C.CustomerID = O.CustomerID JOIN Products P ON O.ProductName = P.ProductName GROUP BY C.Name;
- Output:

	Name	Revenue
1	David Thompson	32.97
2	Emily Brown	58.95
3	James Anderson	38.97
4	Jane Smith	26.97
5	Jessica Lee	17.98
6	John Doe	54.95
7	Kelly Clarkson	47.94
8	Michael Davis	55.94
9	Olivia Martinez	10.99
10	Robert Johnson	11.98
11	Sarah Wilson	56.95
12	William Turner	29.95

7. Write a query to retrieve the customers who have placed at least one order for each product category.

- SELECT C.CustomerID, C.Name FROM Customers C JOIN Orders O ON C.CustomerID = O.CustomerID GROUP BY C.CustomerID, C.Name HAVING COUNT(DISTINCT O.ProductName) = (SELECT COUNT(*) FROM Products);
- Output:

CustomerID	Name

8. Write a query to retrieve the customers who have placed orders on consecutive days.
- SELECT DISTINCT O1.CustomerID FROM Orders O1 JOIN Orders O2 ON O1.CustomerID = O2.CustomerID AND DATEDIFF(DAY, O1.OrderDate, O2.OrderDate) = 1;
- Output:

CustomerID

9. Write a query to retrieve the top 3 products with the highest average quantity ordered.
- SELECT TOP 3 ProductName, AVG(Quantity) AS Avg_Quantity FROM Orders GROUP BY ProductName ORDER BY Avg_Quantity DESC;
- Output:

	ProductName	Avg_Quantity
1	Product E	6
2	Product H	4
3	Product D	3

10. Write a query to calculate the percentage of orders that have a quantity greater than the average quantity.
- SELECT (COUNT*() * 100.0 / (SELECT COUNT(*)* FROM Orders)) AS PercentageAboveAverage FROM Orders WHERE Quantity > (SELECT AVG(Quantity) FROM Orders);
- Output:

	PercentageAboveAverage
1	33.333333333333